

Drug Repurposing Innovation Report

Prompt: Evaluate Montelukast for new respiratory indications and innovation opportunities

Overall Decision & Scores

Decision: GO

Overall Score: 82.2

Clinical Evidence: 75.0

Market Attractiveness: 100.0

Competition: 70.0

Toxicity: 85.0

Patent Risk: 70.0

Regulatory Feasibility: 80.0

Innovation Story

For molecule Montelukast, the system identified high unmet need and attractive market opportunities in: Chronic cough. Based on clinical evidence, market size, toxicity, patent and regulatory analysis, the overall recommendation is: GO (score=82.2).

Recommended Repurposing Indications

Chronic cough

Agent Insights

IQVIA Insights Agent

Summarize market size, growth and competition for Montelukast. Key data: {'molecule': 'Montelukast', 'markets': [{'region': 'US', 'therapy': 'Asthma', 'size': 500000000, 'cagr': 8.5, 'competitors': 5}, {'region': 'India', 'therapy': 'Asthma', 'size': 120000000, 'cagr': 11.0, 'competitors': 2}]}...

EXIM Trade Agent

Summarize API trade flows and supply chain risk for Montelukast. Key data: {'molecule': 'Montelukast', 'trade_flows': [{'api': 'Montelukast', 'from': 'China', 'to': 'India', 'volume_tons': 250}, {'api': 'Montelukast', 'from': 'India', 'to': 'US', 'volume_tons': 100}]}...

Patent Landscape Agent

Summarize patent filings, expiry timeline and FTO risks for Montelukast. Key data: {'molecule': 'Montelukast', 'patents': [{'patent_id': 'US1234567A', 'owner': 'BigPharma Inc', 'expiry_year': 2028, 'fto_risk': 'medium'}, {'patent_id': 'US7654321B', 'owner': 'GenericCo', 'expiry_year': 2026, 'fto_risk': 'low'}]}...

Clinical Trials Agent

Summarize ongoing and completed trials for Montelukast. Highlight trial density and competition. Key data: {'molecule': 'Montelukast', 'trials': [{'id': 'NCT001', 'indication': 'Asthma', 'phase': 'Phase III',

'status': 'Completed'}, {'id': 'NCT002', 'indication': 'COPD', 'phase': 'Phase II', 'status': 'Recruiting'}}]...

Internal Knowledge Agent

Summarize internal strategy and field insights relevant to Montelukast. Key data: {'molecule': 'Montelukast', 'docs': [{'title': 'Field Insights - Respiratory 2024', 'key_points': ['High unmet need in chronic cough', 'Poor adherence in inhaler users']}, {'title': 'Strategy Deck - Emerging Markets', 'key_points': ['India Tier-2 cities growth', 'Price-sensitive but volume rich']}]}...

Web Intelligence Agent

Summarize guidelines, scientific publications and news for Montelukast. Key data: {'molecule': 'Montelukast', 'guidelines': [{'source': 'GINA', 'snippet': 'Montelukast is recommended as add-on therapy in persistent asthma.'}], 'news': [{'headline': 'New studies explore Montelukast in chronic cough', 'url': 'https://example.com/news1'}]}...

Knowledge Graph Agent

Analyze knowledge graph around Montelukast and list related diseases. Key data: {'nodes': [('Montelukast', {'type': 'drug'}), ('Leukotriene receptor antagonist', {'type': 'moa'}), ('Inflammatory pathway', {'type': 'pathway'}), ('Asthma', {'type': 'disease'}), ('Chronic cough', {'type': 'disease'}), ('Allergic rhinitis', {'type': 'disease'})], 'edges': [('Montelukast', 'Leukotriene receptor antagonist', {'type': 'has_moa'}), ('Leukotriene receptor antagonist', 'Inflammatory pa...']

Repurposing Discovery Agent

Suggest repurposing opportunities for Montelukast based on unmet need and KG. Key data: {'candidates': ['Chronic cough'], 'unmet': [{'disease': 'Chronic cough', 'prevalence': 'high', 'unmet_need_score': 90, 'competition_score': 30}, {'disease': 'COPD', 'prevalence': 'high', 'unmet_need_score': 70, 'competition_score': 60}, {'disease': 'Allergic rhinitis', 'prevalence': 'medium', 'unmet_need_score': 50, 'competition_score': 50}]}...

Toxicity Risk Agent

Summarize toxicity and safety profile of Montelukast. Key data: {'molecule': 'Montelukast', 'adverse_events': [{'event': 'Headache', 'frequency': 'very common'}, {'event': 'Neuropsychiatric effects', 'frequency': 'rare'}], 'black_box_warning': False}...

Regulatory Feasibility Agent

Assess regulatory feasibility and fast-track potential for Montelukast. Key data: {'molecule': 'Montelukast', 'approvals': [{'agency': 'FDA', 'indication': 'Asthma', 'year': 1998}, {'agency': 'EMA', 'indication': 'Asthma', 'year': 1999}], 'fast_track_potential': ['Chronic cough'], 'orphan_potential': []]}...

Market Attractiveness Agent

Evaluate commercial attractiveness and revenue potential for Montelukast. Key data: {'molecule': 'Montelukast', 'markets': [{'region': 'US', 'therapy': 'Asthma', 'size': 500000000, 'cagr': 8.5, 'competitors': 5}, {'region': 'India', 'therapy': 'Asthma', 'size': 120000000, 'cagr': 11.0, 'competitors': 2}]}...

Portfolio Optimizer Agent

Portfolio view: For this demo, Montelukast is treated as a high-priority candidate given its unmet need and existing approvals. In full implementation, this agent would rank multiple molecules by

overall innovation score.

Report Generator Agent

Formats synthesized response into a polished PDF report, including charts, tables, and references. In this architecture, this agent is used indirectly via MasterAgent.